SCIENTIST GUIDE

RatioBos 1.0.0

From the *factorial animal* to the certified ration

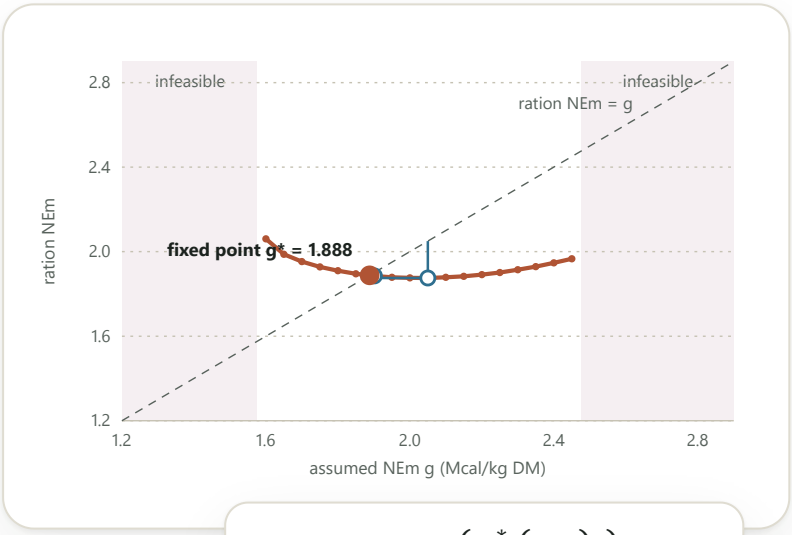
Least-cost beef cattle formulation: NASEM (2016) equations, iterative energy–intake adjustment, linear programming with a duality certificate, IPCC (2019) environmental footprint and validation against published diets.

1st edition · October 2026 · covers RatioBos 1.0.0

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Companion to the Producer Guide.



$$g_*:\ \mathrm{ENm}_{\mathrm{dieta}}\left(\mathbf{x}^*\left(g_*\right)\right) = g_*$$

ANIMAL

NASEM (2016)

California net energy and metabolizable protein

METHOD

Simplex + fixed point

assumed and ration NEm agree

VALIDATION

19 / 19 · 0.5 %

diets matched or cheaper · observed NE error

TEST

500 / 500

problems identical to HiGHS (SciPy)

BEFORE READING

Scope, notation and citation

Who this guide is for, what it assumes of the reader and how it is organised.

This guide documents RatioBos's method in the detail needed to evaluate, teach and reproduce it: every animal equation, the construction of the linear model, the algorithm that makes the ration's energy match the one assumed to predict intake, the environmental footprint and the validation against published experiments. It assumes knowledge of ruminant nutrition and a basic grasp of linear programming.

Every number comes from the app with the same case as the Producer Guide: **300 zebu × European cross bulls at 380 kg**, 1.35 kg/d target, Jalisco in the dry season, implanted and fed monensin (30 mg/kg DM). In the app, the **scientist view** (button in the top bar) shows everything described here.

Boxes

≡ **DEFINITION**

A concept with its exact meaning.

∴ **RESULT**

A property the program relies on, with its practical consequence.

◆ **IN THE CASE**

The case numbers, exactly as the app computes them.

! **CAUTION**

An assumption or limitation that can change a conclusion.

Notation

SYMBOL	MEANING	UNIT
PV · SBW	live weight · shrunk body weight (0.96 LW)	kg
EQSBW	equivalent weight: SBW × SRW / FSBW	kg
SWG · GDP	shrunk weight gain · daily gain	kg/d
CMS	dry matter intake (DMI)	kg/d
ENm · ENg	net energy for maintenance · for gain	Mcal/kg MS
ER	retained energy (daily NEg required)	Mcal/d
PM · PCM · PNDR	metabolizable · microbial crude · rumen-undegradable protein	g
x_i	inclusion of ingredient i	% of DM
c_i	ingredient price on a DM basis	$/kg MS
g	assumed ration NEm	Mcal/kg MS

How to cite

" **SOFTWARE**

Mojica-Zárate, H. T., & Barrera-Guzmán, L. Á. (2026). *RatioBos: Guided least-cost diet formulation for beef cattle* (Version 1.0.0) [Software]. Zenodo. https://doi.org/10.5281/zenodo.23197134

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THIS GUIDE (APA 7TH ED.)

Mojica-Zárate, H. T., & Barrera-Guzmán, L. Á. (2026). *RatioBos: Scientist Guide* (1st ed.). https://hectortecumshe-ai.github.io/RatioBos/instructivo-cientifico.html

BLOCKS 1 TO 4

The animal model

Factorial requirements from the California net energy system (Lofgreen & Garrett, 1968) with the NRC (1996, 2000) and NASEM (2016) equations.

M.1 Equivalent weight

Gain composition depends on how close the animal is to its target-fatness weight. NASEM handles this by scaling the animal's weight to a reference animal with 478 kg shrunk weight at 28 % carcass fat (SRW):

$$\mathrm{SBW} = 0.96\ \mathrm{PV}, \qquad \mathrm{EQSBW} = \mathrm{SBW} \cdot \frac{\mathrm{SRW}}{\mathrm{FSBW}} \tag{1}$$

where FSBW is 0.96 times the final reference weight in Block 3 (or the mature weight for heifers).

◆ **IN THE CASE**

SBW = 0.96 × 380 = 364.8 kg; FSBW = 0.96 × 620 = 595.2 kg; EQSBW = 364.8 × 478 / 595.2 = **293.0 kg**.

M.2 Maintenance

$$\mathrm{ENm}_{\mathrm{req}} = \left(0.077\, f_{\mathrm{raza}}\, f_{\mathrm{sexo}}\, f_{\mathrm{lact}} + a_2\right) \mathrm{SBW}^{0.75} \cdot f_{\mathrm{jadeo}} \cdot f_{\mathrm{act}} \tag{2}$$

FACTOR	VALUES	SOURCE
f_{raza}	British and Continental 1.00; zebu 0.90; cross 0.95; Criollo 1.00; dairy 1.20	NRC (1996)
f_{sexo}	steer and heifer 1.00; bull 1.15	NRC (1996)
f_{lact}	1.20 in lactating cows	NRC (1996)
a_2	0.0007 (20 − $T_{previous}$): acclimation to the previous month's temperature	NRC (1996)
f_{jadeo}	1.07 with rapid panting (effective THI 79–83); 1.18 open-mouth (≥ 84)	NRC (1996)
f_{act}	feedlot 1.00; semi-confined 1.05; flat grazing 1.10; hill grazing 1.20	NRC (1996)

THI (NRC, 1971) is 0.81 T + RH/100 (T − 14.4) + 46.4. The **effective THI** credits shade (sheet −2 points, ample shade −4, dome with fans −6) and biotype adaptation (zebu +3 points; crosses and Criollo +1.5; dairy −1), and sets the panting factor.

◆ **IN THE CASE**

Previous month at 17 °C: a_2 = 0.0021. NEm_{req} = (0.077 × 0.95 × 1.15 + 0.0021) × $364.8^{0.75}$ = **7.20 Mcal/d**. No panting (effective THI 64.1), feedlot.

M.3 Intake

Expected intake depends on the ration's NEm (NRC, 1996). For growing cattle:

$$\mathrm{CMS} = \mathrm{SBW}^{0.75} \cdot \frac{0.2435\,g - 0.0466\,g^2 - k}{g} \cdot \prod_m F_m \qquad (3)$$

with k = 0.1128 for yearlings and 0.0869 for calves, and g the ration NEm in Mcal/kg DM. For cows: DMI = $\mathrm{SBW}^{0.75}$ (0.04997 g^2 + 0.03840)/g (0.04631 in lactation) + 0.2 × milk. The factors F are: body fat by EQSBW

(from 1.00 to 0.73 between 350 and 550 kg), no implant 0.94, ionophore 0.97, temperature (0.65 to 1.16) and mud (–1 % per cm). The user can replace predicted intake with observed intake.

M.4 Energy and protein of gain

$$\begin{aligned} \mathrm{ER} &= 0.0635\,(0.891\,\mathrm{EQSBW})^{0.75}\,(0.956\,\mathrm{SWG})^{1.097} \\ \mathrm{PN}_g &= \mathrm{SWG}\,(268 - 29.4\,\mathrm{ER}/\mathrm{SWG}) \\ \mathrm{PM}_g &= \mathrm{PN}_g / e, \quad e = 0.834 - 0.00114\,\mathrm{EQSBW} \quad (\geq 0.492) \\ \mathrm{PM}_m &= 3.8\,\mathrm{SBW}^{0.75} \end{aligned} \tag{4}$$

◆ **IN THE CASE**

With SWG = 1.35 kg/d: RE = **5.46 Mcal/d**; NP_g = 201.4 g/d; e = 0.500; MP_g = 402.8 g/d; MP_m = 317.2 g/d; total MP = **720 g/d**.

M.5 Metabolizable protein supply

Microbial synthesis is estimated with the Galyean and Tedeschi (2014) equation, adopted by NASEM (2016), from TDN intake; added fat is excluded because it does not feed the microbes:

$$\mathrm{PCM} = 42.73 + 0.087\,\mathrm{TND}_{\mathrm{cons}}, \qquad \mathrm{PM} = 0.64\,\mathrm{PCM} + 0.80\,\mathrm{PNDR} \tag{5}$$

The degradable protein must also cover microbial protein (RDP − MCP balance ≥ 0); otherwise the MP supply is reduced accordingly. The constant 42.73 g/d is divided by intake to express it per kilo of DM, so the constraint stays linear in the inclusions.

M.6 Pregnancy, lactation and minerals

Pregnancy (NRC, 1996), with t days pregnant and BW_b the birth weight:

$$\begin{aligned} \mathrm{EM}_y &= \mathrm{PN}\left(0.4504 - 0.000766\,t\right) e^{\,(0.03233 - 0.0000275t)\,t} / 1000 \\ \mathrm{PM}_y &= 6.25\,\mathrm{PN}\left(0.001669 - 0.00000211\,t\right) e^{\,(0.0278 - 0.0000176t)\,t} / 0.65 \end{aligned} \tag{6}$$

ME_y is converted to NEm with k_m = ration NEm/ME. Lactation: NEl = milk × (0.3512 + 0.0962 × fat %) and MP_l = milk × protein % × 10 / 0.65. Calcium and phosphorus are factorial: maintenance 15.4 and 16 mg/kg

SBW, growth 7.1 % and 3.9 % of retained protein, plus late pregnancy and milk, with 50 % (Ca) and 68 % (P) absorption.

> **!** **CAUTION · PRACTICAL LIMITS**
>
> Forage, physically effective NDF, fat, sulfur, sodium and maximum crude protein do not come from equations: they are practical limits per animal type (e.g. 8–15 % forage and peNDF ≥ 7 % in finishing). The sulfur maximum is reduced by the sulfur supplied by water (3.5–5 L per kg DM depending on temperature).

BLOCKS 5 TO 7

Formulation: linear program and fixed point

Least cost comes from linear programming; the dependence between ration energy and intake is resolved with a fixed point around the linear program.

F.1 The linear program

The variables are the inclusions x_i in % of DM. For a given intake C and per-kilo requirements, RatioBos solves:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \textstyle\sum_i c_i x_i \\ \text{subject to} \quad & \textstyle\sum_i x_i = 100 \\ & \textstyle\sum_i a_{ij} x_i \geq 100\left(b_j^- - k_j\left(C\right)\right), \quad \sum_i a_{ij} x_i \leq 100\left(b_j^+ - k_j\left(C\right)\right) \\ & l_i \leq x_i \leq u_i \end{aligned} \tag{7}$$

where a_{ij} is the contribution of 1 kg DM of ingredient i to nutrient j and $k_j(C)$ is the constant term of microbial synthesis ($0.64 \times 42.73/C$ for MP and $-42.73/C$ for the RDP balance; zero otherwise). The per-ingredient contributions of the derived constraints are:

CONSTRAINT	CONTRIBUTION OF 1 KG DM OF THE INGREDIENT
Metabolizable protein (g/kg)	$0.64 \times 0.087 \times 10\ \mathrm{TND_i} + 0.80 \times 10\ \mathrm{PC_i}\ (1 - \mathrm{PDR_i}/100)$
RDP − MCP balance (g/kg)	$10\ \mathrm{PC_i}\ \mathrm{PDR_i}/100 - 0.087 \times 10\ \mathrm{TND_i}$
Physically effective NDF (%)	$\mathrm{FDN_i} \times \mathrm{pef_i}$
Forage (%)	100 for dry forage, silage or fresh forage; 0 otherwise

For added fat $\mathrm{TDN_i}$ is taken as zero in the first two rows. Each row is scaled by its largest absolute coefficient before solving.

F.2 Simplex and optimality certificate

METHOD	**Two-phase tableau simplex**	PIVOTING	**Bland's rule (anti-cycling)**
ARITHMETIC	**Double precision (IEEE 754)**	INFEASIBILITY	**Artificial sum > 10^{-7}**
CERTIFICATE	**Strong duality, gap < 10^{-7}**	DIAGNOSIS	**One-constraint-at-a-time relaxation**

∴ **RESULT · STRONG DUALITY**

If $\mathbf{x}$ is primal feasible, $\mathbf{y}$ is dual feasible and $\mathbf{c}^{\mathrm{T}}\mathbf{x} = \mathbf{b}^{\mathrm{T}}\mathbf{y}$, both are optimal (Dantzig, 1963). RatioBos checks all three conditions on the numerical solution before displaying it; if one fails, it says so.

◆ **IN THE CASE**

12 variables, 21 constraints, 39 pivots. Primal violation 4.3×10^{-14}, dual violation 1.5×10^{-13}, relative gap $\mathbf{2.0 \times 10^{-15}}$. Binding constraints: NEg ≥ 1.254 Mcal/kg (shadow price \$2.39 per Mcal/kg), MP ≥ 92.8 g/kg (\$0.0095 per g/kg), RDP balance ≥ 0, Ca ≥ 0.514 %, Na ≥ 0.08 % and premix fixed at 0.3 %.

The **shadow price** reads as the change in cost per kilo of DM when the requirement rises by one unit: demanding 0.01 Mcal/kg more NEg raises the ration cost by \$23.9 per tonne of DM. The **reduced cost** of an unused ingredient is how much its price per kilo of DM must fall for it to enter; converted to an as-fed basis it gives the "entry price" the producer sees.

F.3 The energy–intake adjustment

The linear program needs intake C and the NEg required per kilo, but both depend on the ration NEm, which is not yet known. For an assumed NEm g, intake comes from (3) and the NEg required per kilo is

$$\mathrm{ENg}_{\mathrm{req}}\left(g\right) = \frac{\mathrm{ER}}{C\left(g\right) - \mathrm{ENm}_{\mathrm{req}} \,/\, \left(g\, f_{\mathrm{ion}}\right)} \tag{8}$$

(feed spent on maintenance is not available for gain). With these data (7) is solved, giving a ration with its own NEm, $NEm_{diet}(g)$. The consistent solution is a **fixed point**: $f(g) = NEm_{diet}(g) - g = 0$. RatioBos searches for it as follows:

1. **Damped direct steps**: $g \leftarrow g + 0.85\ f(g)$, up to 7 evaluations, stopping when $|f| < 0.002$ Mcal/kg.
2. If it does not converge, a **sweep** of g from 0.8 to 2.9 Mcal/kg in 0.1 steps, and **bisection** (30 steps, tolerance 0.0005) on the first pair of feasible rations with a sign change ($f \geq 0 \rightarrow f < 0$).
3. The ration on the **conservative** side ($f \geq 0$) is kept: its true NEm is at least the assumed one, so it meets the true requirement. With no sign change, the feasible one with $f \geq 0$ closest to zero is used.

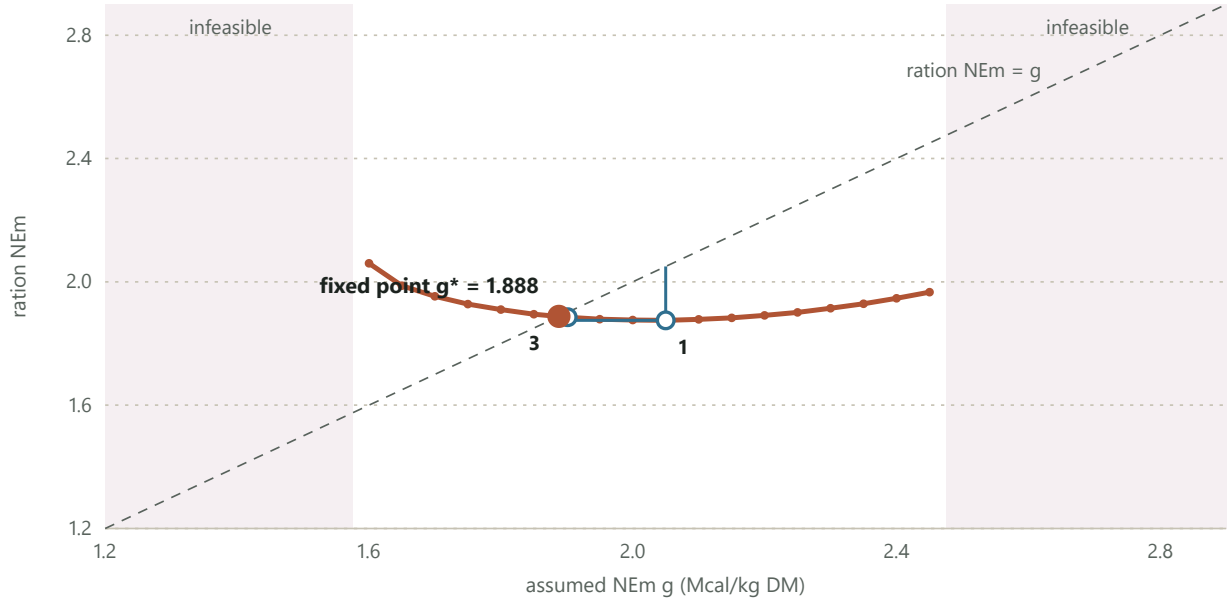


FIGURE F.1 **The case fixed point.** Terracotta curve: NEm of the least-cost ration for each assumed NEm; dashed line: identity. In blue, the three iterations the app made from g = 2.05. Outside 1.6–2.45 Mcal/kg the linear program has no solution.

◆ **IN THE CASE**

Starting from $g = 2.05$: the resulting ration has 1.875 Mcal/kg; with $g = 1.901$ it gives 1.885; with $g = 1.888$ it gives 1.8875 ($|f| = 0.0003$). Three linear programs sufficed. Final intake 7.75 kg DM/d; NEg required 1.254 Mcal/kg.

! **CAUTION · WHY PLAIN ITERATION IS NOT ENOUGH**

With low gain targets, the least-cost linear program can return rations with NEm far below the assumed value, and with low assumed NEm the problem turns infeasible (intake falls and the NEg required per kilo soars). Plain iteration oscillates or stalls in the infeasible region; the sweep with bisection finds the crossing if it exists. Energy-free minerals (limestone, phosphate) are capped at 2 % of DM to stop them acting as cheap energy "diluents".

F.4 Maximum feasible gain

If (7) has no solution for the target, bisection (14 steps between 0.1 kg/d and the target) finds the highest gain for which the F.3 procedure yields a feasible ration. It is the gain behind the "Use … kg/d and formulate" button, and the one the program and the regional comparison use when the climate does not allow the target.

BLOCKS 7 TO 9

D Predicted performance and economics

From ration to gain, from one day to the whole cycle and from the cycle to profit.

D.1 Energy- and protein-allowable gain

With the formulated ration, feed for maintenance is FM = $NEm_{req}/(NEm_{diet} \cdot f_{ion})$ and energy available for gain is RE_{av} = (C − FM) NEg_{diet}. **Energy-allowable gain** is the exact inverse of (4), solved by bisection (60 steps). **Protein-allowable gain** solves $NP_g(SWG) = (MP_{supplied} - MP_m) \cdot e$. Predicted gain is the smaller of the two, and the app reports which one limits.

◆ **IN THE CASE**

FM = 3.40 kg DM/d; C − FM = 4.35 kg DM/d; RE_{av} = 5.455 Mcal/d; energy-allowable gain 1.34996 kg/d and protein-allowable 1.35001 kg/d: **both constraints bind at once**, the mark of a well-balanced ration.

D.2 Program simulation

For each phase a ration is formulated at the phase's mid-weight, gain target (or the maximum feasible, found by bisection) and forage range (in feedlot receiving, also peNDF ≥ 12 %), then growth is simulated **day by day**: each day expected intake (with the microbial MP term adjusted to that intake) and the gain that ration allows at that day's weight are recomputed, until the phase end weight is reached. Days, DM, cost, methane and nitrogen are accumulated. Profit includes purchase, feed, other daily costs and sale; the break-even price is the sale price that makes it zero.

D.3 Economic optimum gain, regions and sensitivity

The dashboard formulates the ration over a grid of gain targets and computes, for each, the daily margin, ADG × sale price − DMI × cost − other costs, and the feed cost per kilo gained. The economic optimum is the highest-margin target; in the case, 1.40 kg/d ($36.44 per head per day versus $34.30 at 1.35 kg/d). The regional comparison repeats procedure F.3 with each region's seasonal climate and price factors; the sensitivity analysis moves the price of the ingredient with the largest cost share between 70 % and 130 % and reformulates at each point.

BLOCK 10

Environmental footprint and cost–footprint frontier

Per-head daily inventory with IPCC (2019) methods and AR6 warming potentials, and a bi-objective formulation via an internal carbon price.

DEFINITION · SUSTAINABILITY

RatioBos follows FAO (2014, 2018): a system is sustainable when it meets present needs without compromising the ability of future generations to meet theirs, across its environmental, economic and social dimensions. Hence the block does not minimise footprint in isolation, but shows the economic cost of reducing it.

S.1 Greenhouse gases

$$
\begin{aligned}
\mathrm{CH}_{4,\,\mathrm{ent}} &= C \cdot 18.45 \cdot \frac{Y_m}{100} / 55.65 \quad (\mathrm{kg/d}) \\
\mathrm{CH}_{4,\,\mathrm{est}} &= \mathrm{VS} \cdot B_0 \cdot 0.67 \cdot \mathrm{MCF}, \quad \mathrm{VS} = C\left(1 - \mathrm{DE} + 0.04\right)\left(1 - 0.08\right) \\
N_2O &= N_{\mathrm{ex}}\left[\mathrm{EF}_3 + \mathrm{Frac}_{\mathrm{gas}} \cdot 0.010 + \mathrm{Frac}_{\mathrm{leach}} \cdot 0.011\right] \cdot 44 / 28 \\
\mathrm{CO}_2e &= \textstyle\sum_i C_i e_i + 27\left(\mathrm{CH}_{4,\,\mathrm{ent}} + \mathrm{CH}_{4,\,\mathrm{est}}\right) + 273\, N_2O
\end{aligned}
\tag{9}
$$

The ration $\mathbf{Y_m}$ is the weighted mean of each ingredient's value under the IPCC (2019, table 10.12) categories: concentrates 3.0 %; forages 4.0, 6.3 or 7.0 % by TDN (≥ 72, ≥ 62 or lower); minerals, additives and urea 0. B_0 = 0.18 m³/kg VS; MCF of 1.0, 1.5 or 2.0 % in a dry lot by temperature (< 15, 15–25, > 25 °C) and 0.47 % on pasture. EF_3 = 0.02 in feedlots and 0.004 on pasture; $\mathrm{Frac}_{\mathrm{gas}}$ = 0.30 and 0.21; $\mathrm{Frac}_{\mathrm{leach}}$ = 0.24 only in humid regions. N_{ex} = N intake − N retained (NP_g/6.25 in growing cattle). e_i is the footprint of producing 1 kg DM of ingredient i (crop, fertiliser, processing and transport, excluding land-use change), with indicative values within the GFLI and FAO-GLEAM ranges and economic allocation for by-products; all are editable in the scientist view.

◆ **IN THE CASE**

Y_m = 3.38 %; enteric CH_4 86.9 g/d; N intake 162.3 g/d, retained 32.2 g/d, excreted 130.1 g/d. CO_2e: feed 3.08, enteric 2.35, N_2O 1.43 and manure CH_4 0.09 kg/d; total **6.95 kg CO_2e/d**, 5.15 kg per kg gained.

S.2 Water, land and manure

INDICATOR	COMPUTATION	SOURCE
Drinking water	3.5 L/kg DM (5 L above 25 °C)	NASEM (2016)
Feed water footprint	Σ C_i × green + blue water per kg DM	Mekonnen y Hoekstra (2011)
Grey water	Leached N / 10 mg N/L	Mekonnen y Hoekstra (2011)
Land for feed	Σ C_i × m²·yr per kg DM (mean yields)	SIAP (2024)
Land for manure	$\max(N_{ex}(1 - Frac_{gas})/170, P_{ex}/25)$ ha·yr	170 kg N and 25 kg P/ha limits
Ammonia	$N_{ex} \times Frac_{gas} \times 17/14$	IPCC (2019)

S.3 Formulation with an internal carbon price

To trace the cost–footprint frontier, the same F.3 procedure is solved with an extended cost $c_i + w \cdot CO_2e_i/1000$, where CO_2e_i (kg per kg DM) adds the feed footprint, the enteric methane attributable through its Y_m, and its manure methane and N_2O, and *w* runs through 0, 1,000, 2,000, 3,500, 5,000, 8,000, 12,000 and 20,000 $/t CO_2e. Every solution is Pareto-efficient (weighting method). The suggested **balance** point is the one that maximises CO_2e reduction per extra peso among those cutting the footprint by at least 2 %.

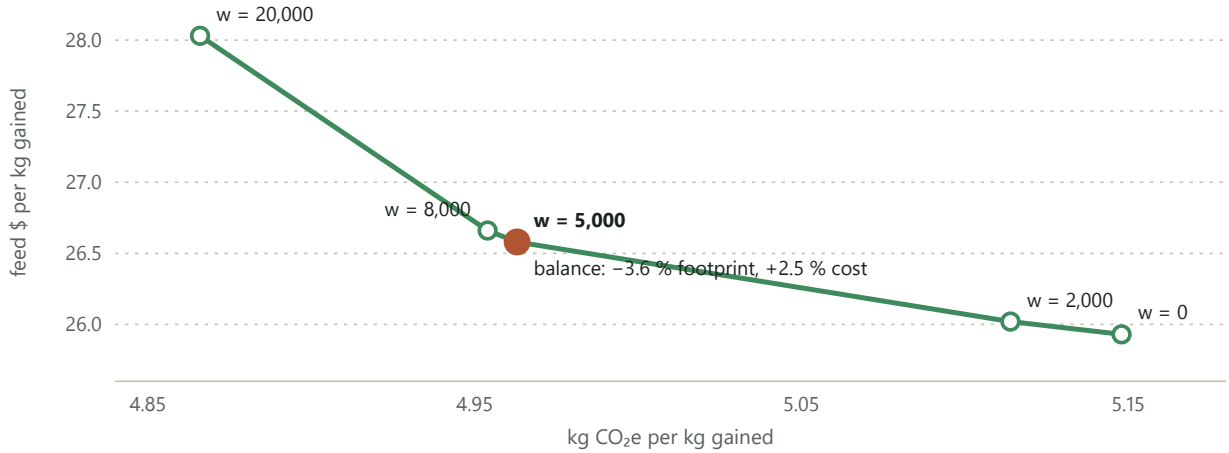


FIGURE S.1 **Cost–footprint frontier for the case.** Each point is an optimal ration under an internal carbon price w (\$/t CO_2e). At w = 1,000, 3,500 and 12,000 the ration does not change from the previous point. Each tonne avoided at the balance point costs ≈ \$3,500.

BLOCK ✓

Validation against published diets

V

Five independent tests: composition, optimiser, observed net energy, economics and metabolizable protein; plus a solver comparison with HiGHS.

V.1 The experiments

The formulas of 19 diets from 9 papers published 2017–2025 were entered as published, with 27 animal groups with measured performance: crossbred bull feedlots in Sinaloa, high-risk calves in Zacatecas, and Holstein calves and steers from Zinn's group at El Centro, California. Ingredients missing from the library were added with the composition each paper reports.

PAPER	SITE	ANIMALS	DIETS	GROUPS
Estrada-Angulo et al. (2022)	Culiacán, Sinaloa	Crossbred bulls, finishing	1	2
Plascencia et al. (2022)	El Centro, California	Holstein calves, growing	2	2
Ramos-Aviña et al. (2018)	El Centro, California	Holstein steers, d 1–112	2	2
Castro-Pérez et al. (2020)	Culiacán, Sinaloa	Bulls in tropical climate (shade)	3	4
Rodríguez-Cordero et al. (2025)	Zacatecas	High-risk bull calves	1	5
González-Vizcarra et al. (2017)	El Centro, California	Steers, dry-rolled vs flaked corn	2	4
Montano et al. (2023)	El Centro, California	Holstein calves, DDGS	4	4
Salinas-Chavira et al. (2024)	El Centro, California	Holstein calves, MP and methionine	2	2
Carvalho et al. (2022)	El Centro, California	Holstein steers, MP and virginiamycin	2	2
Total			**19**	**27**

V.2 Test 1 · computed composition

The app analyses each formula with its own library and compares the result with the composition reported by the authors. Mean absolute errors are 3.0 % for net energy (38 values), 2.1 % for crude protein, 4.4 % for fibre and fat and 7.8 % for minerals.

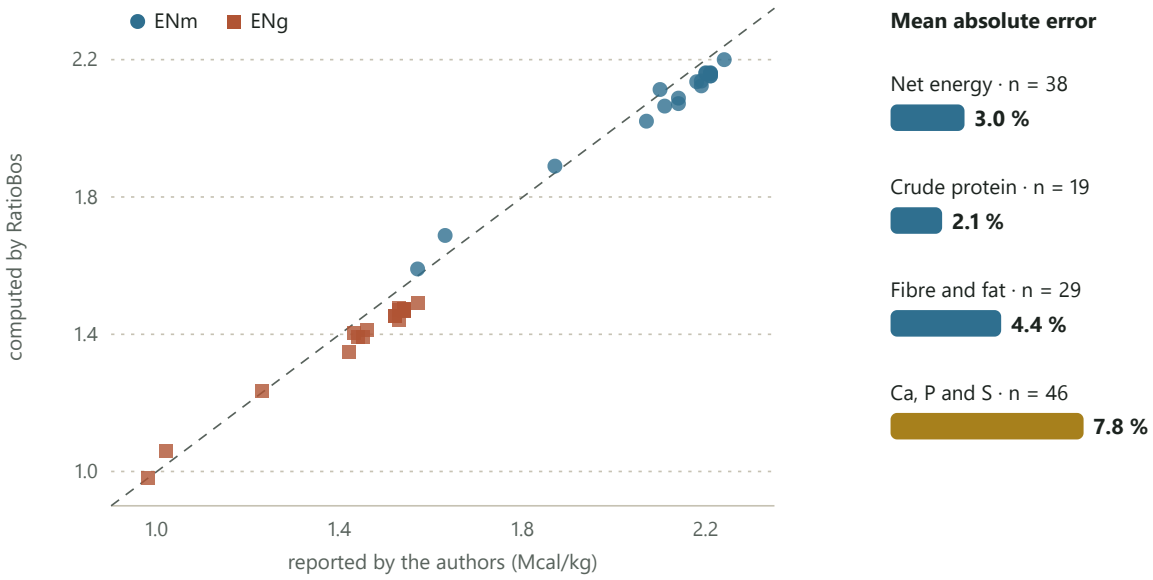


FIGURE V.1 **Reported versus computed composition.** Left: net energy of the 19 diets (dashed line = equality). Right: mean absolute error by nutrient group.

> **!** **CAUTION · BIAS IN TABULAR ENERGY**
>
> On steam-flaked corn diets the app gives 2–3 % less NEm and 3–6 % less NEg than the authors. This is not a computing error: the papers use higher tabular energy values for that grain (≈ 2.4 Mcal NEm/kg) than the Oklahoma State table shipped with the app (2.29). On the higher-forage diets (the step-up diets of Castro-Pérez et al., 2020, and the receiving diet of Rodríguez-Cordero et al., 2025) the bias vanishes or reverses. If you know your ingredient's value, edit it in Block 5.

V.3 Test 2 · the optimiser

For each published diet, its own nutrient supply was set as the requirement (minimums for energy, protein and minerals; maximums for NDF, fat and sulfur), and the optimiser chose among the same ingredients, with premix and additives fixed. If the optimiser works, it must return a ration at least as cheap as the published one. Result: **19 of 19**, all with a valid duality certificate.

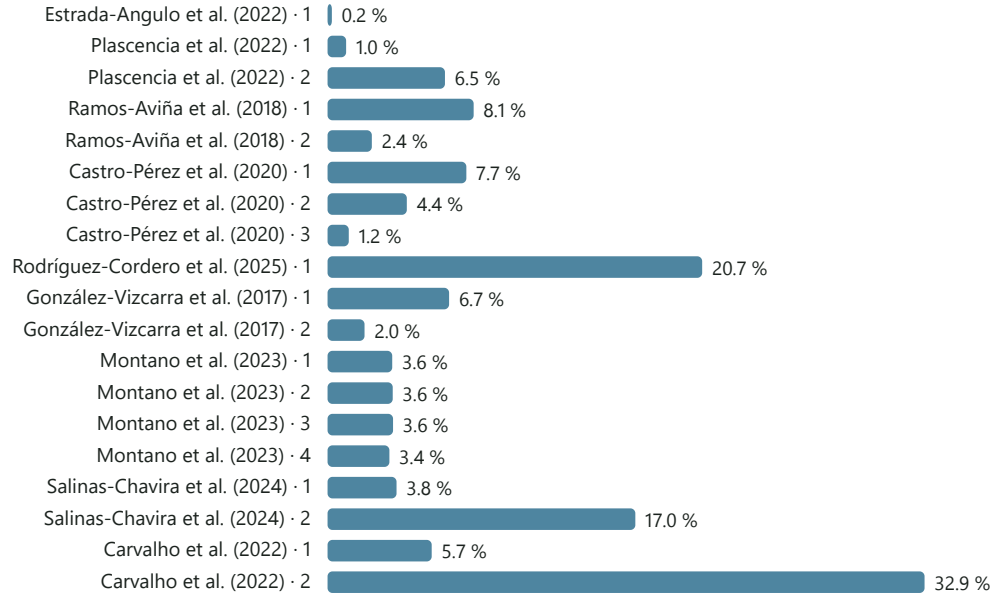


FIGURE V.2 **Optimiser test.** For all 19 diets the app found an equally nutritious, cheaper ration with the same ingredients (costs at the app's example prices).

V.4 Test 3 · observed net energy

Papers from Zinn's group report the diet net energy "observed" from animal performance (Zinn & Shen, 1998). RatioBos repeats that calculation with the published weights, gain and intake, solving for NEm the quadratic

$$C = \frac{e_m W^{0.75}}{\mathrm{ENm}} + \frac{e_g W^{0.75} \mathrm{GDP}^{1.097}}{0.877\,\mathrm{ENm} - 0.41} \tag{10}$$

with W the mean weight and each paper's e_m and e_g coefficients. The mean absolute error against the published values is **0.5 %** (27 groups, NEm and NEg): the app reproduces the authors' method.

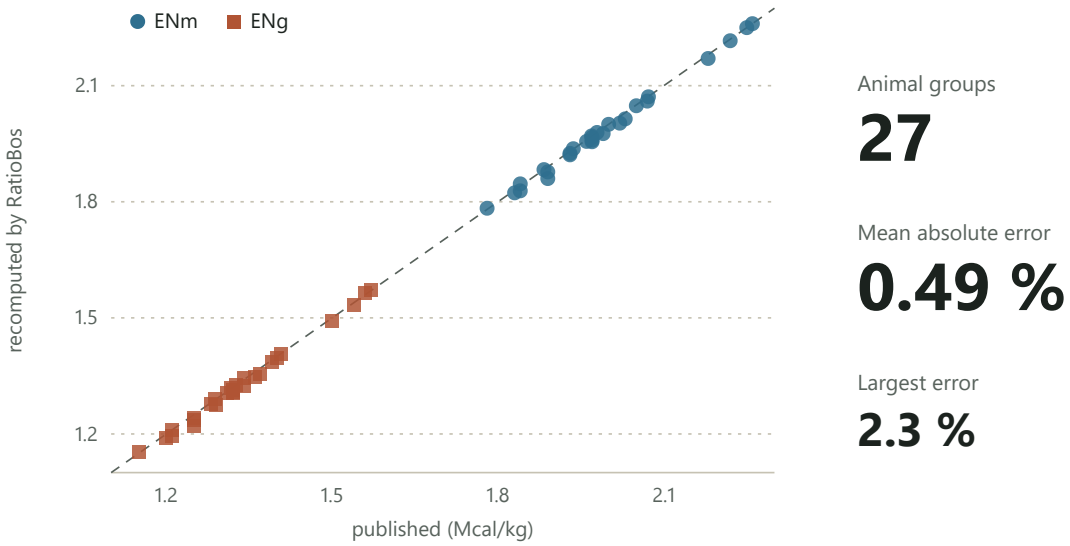


FIGURE V.3 **Observed net energy.** From the published weights, gain and intake, the app reproduces the "observed" NEm and NEg reported by the authors (Zinn & Shen, 1998).

V.5 Tests 4 and 5 · economics and metabolizable protein

With the prices in Rodríguez-Cordero et al. (2025), the cost per kilo gained computed by the app differs by 0.2 % from the published value. With the Montano et al. (2023) diets, MP supplied as a percentage of MP required, computed with RatioBos equations, averages 3.5 points below the authors' estimate (89–98 % versus 93–100 %), with the same ranking across treatments.

V.6 The solver against HiGHS

500 real app problems were generated with a fixed seed (all 5 stages, 6 biotypes, 9 regions and 3 seasons; random library ingredients; prices from −50 % to +50 %; assumed NEm 1.2–2.3 Mcal/kg) and also solved with HiGHS (SciPy 1.18). All 500 agreed: in the 288 feasible ones cost differs by at most 4×10^{-14} in relative terms, and the 212 infeasible ones are infeasible for both. The scripts are in the repository's `pruebas` folder.

SCIENTIST VIEW

Reproducibility and publication

What the scientist view shows, how to export the model and how to verify it outside the app.

R.1 What the scientist view adds

BLOCK	ADDITIONAL CONTENT
3	All 15 nutrients and indicators, intake factors, assumed NEm and target-fatness weight (SRW).
5	Full composition of each ingredient and its footprint factors (CO_2e, water, land).
7	Model card (variables, constraints, pivots, certificate), energy-adjustment iteration table, shadow prices, reduced costs, energy- and protein-allowable gain, methane and nitrogen, and JSON export of the model.
✓	Each published diet: reported versus computed composition, optimiser test, observed energy, economics and MP.
§	Full theoretical framework with APA references.

R.2 Independent verification

The **⬇ Model and solution (JSON)** button in Block 7 saves the animal, conditions, intake, requirements, iterations, ingredients with their composition, price and limits, and the complete solution. The script below rebuilds linear model (7) from that file, solves it with HiGHS and compares. With the case: identical formula and zero relative cost difference.

pruebas/verificar_ratiobos.py · Python ≥ 3.9 · SciPy ≥ 1.9

```
"""Verifica con HiGHS (SciPy) una ración exportada desde RatioBos (vista científica,
botón «Modelo y solución (JSON)»). Reconstruye el modelo lineal a partir de los
ingredientes, los requerimientos y el consumo, lo resuelve y compara el costo y
la fórmula con los de la app.   Uso:  python verificar_ratiobos.py archivo_modelo.json"
import json
import sys

import numpy as np
from scipy.optimize import linprog

m = json.load(open(sys.argv[1] if len(sys.argv) > 1 else "ratiobos_modelo.json", encodi
ings, req, dmi = m["ingredientes"], m["requerimientos"], m["consumoMS"]
FORRAJE = {"for", "ens"}
MCP_A, MCP_B, MP_MCP, D_RUP = 42.73, 0.087, 0.64, 0.80      # Galyean y Tedeschi (2014)


def aporte(ing, k):
    """Aporte de 1 kg de MS del ingrediente al nutriente k (misma definición que la app
    c = ing["comp"]
    tnd = 0 if ing["cat"] == "gras" else c.get("TND", 0)          # la grasa no aliment
    pc, pdr = c.get("PC", 0), c.get("PDR", 0)
    if k == "PM":
        return MP_MCP * MCP_B * tnd * 10 + pc * 10 * (1 - pdr / 100) * D_RUP
    if k == "PDRb":
        return pc * 10 * pdr / 100 - MCP_B * tnd * 10
    if k == "PDR":
        return pc * pdr / 100
    if k == "FDNef":
        return c.get("FDN", 0) * c.get("pef", 0)
    if k == "For":
        return 100 if ing["cat"] in FORRAJE else 0
    return c.get(k, 0)


def constante(k):
    """Término constante de la síntesis microbiana por kg de MS (depende del consumo)."
    return {"PM": MP_MCP * MCP_A / dmi, "PDRb": -MCP_A / dmi}.get(k, 0)


precio = np.array([i["precio"] / max(i["comp"]["MS"] / 100, 0.01) for i in ings])   # $
A_ub, b_ub = [], []
for k, (lo, hi) in req.items():
    fila = np.array([aporte(i, k) for i in ings])
```

```
    if lo is not None:
        A_ub.append(-fila); b_ub.append(-100 * (lo - constante(k)))
    if hi is not None:
        A_ub.append(fila); b_ub.append(100 * (hi - constante(k)))
cotas = [(i.get("min") or 0, i["max"] if i.get("max") not in (None, "") else 100) for i
r = linprog(precio, A_ub=np.array(A_ub), b_ub=b_ub, A_eq=[np.ones(len(ings))], b_eq=[10
            bounds=cotas, method="highs")
if r.status != 0:
    sys.exit("HiGHS: " + r.message)

app = {f["id"]: f["pct"] for f in m["solucion"]["formula"]}
costo_app, costo_highs = m["solucion"]["costoKgMS"], r.fun / 100
print(f"{'Ingrediente':38s} {'RatioBos':>9s} {'HiGHS':>9s}")
for i, x in zip(ings, r.x):
    if x > 1e-6 or app.get(i["id"], 0) > 1e-6:
        print(f"{i['nombre'][:38]:38s} {app.get(i['id'], 0):9.3f} {x:9.3f}")
print(f"\nCosto por kg de MS · RatioBos: {costo_app:.6f}  HiGHS: {costo_highs:.6f}")
print(f"Diferencia relativa: {abs(costo_app - costo_highs) / costo_highs:.2e}")
```

R.3 Text for Materials and methods

A base paragraph to describe the use of RatioBos in a paper. Adjust the bracketed data:

Rations were formulated at least cost with RatioBos version 1.0.0 (Mojica-Zárate & Barrera-Guzmán, 2026). Net energy, metabolizable protein, calcium and phosphorus requirements were computed with the NASEM (2016) equations for [380 kg zebu × European crossbred bulls with a 1.35 kg/d target gain], including the NRC (1996) adjustments for biotype, sex, thermal acclimation, temperature-humidity index and activity, and the microbial protein synthesis of Galyean and Tedeschi (2014). Intake was predicted with the NRC (1996) equations [or: fixed at observed intake]. The dependence between ration energy and intake was resolved by iterating until assumed and ration NEm differed by less than 0.002 Mcal/kg DM. Each linear program was solved with the two-phase simplex method under Bland's rule, and optimality was verified by strong duality (relative primal–dual gap $< 10^{-7}$). Ingredient composition was taken from Beck et al. (2024) [or: from laboratory analyses], with inclusion limits of [describe]. The greenhouse-gas footprint was estimated with the IPCC (2019) Tier 2 method and AR6 100-year global warming potentials. The complete model and solution were exported as JSON and are provided as supplementary material.

SCOPE

Assumptions and limitations

What the model does not do, and when to distrust a number.

- **Tabular energy.** Ingredient energy is tabular and constant; it is not adjusted for intake level, fine processing or associative effects. Grain processing matters a lot (dry-rolled versus flaked ≈ 15 %).
- **Protein.** MP uses microbial synthesis from TDN (NASEM level 1); individual amino acids and passage rates are not modelled.
- **Intake.** The NRC (1996) intake equations explain a limited share of pen-to-pen variation; if you know actual intake, enter it.
- **Climate.** Regional normals are approximate and the heat response is reduced to maintenance and intake factors; there is no heat-balance model.
- **Prices.** Regional prices are an example with estimated factors; they do not replace the user's quotes.
- **Footprint.** Per-ingredient factors are indicative, exclude land-use change and do not amount to a certified life-cycle assessment.
- **Linearity.** The program is linear in the inclusions; nonlinear effects (fat on fibre digestion, for instance) are handled with limits, not in the objective.

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RatioBos · Scientist Guide, 1st edition, October 2026. Covers RatioBos 1.0.0. Every number comes from the app with the Producer Guide case; the figures are drawn from those data.

Decision-support tool: results should be reviewed by a qualified animal nutritionist. © 2026 Héctor Tecumshé Mojica-Zárate. The software is distributed under the MIT License.